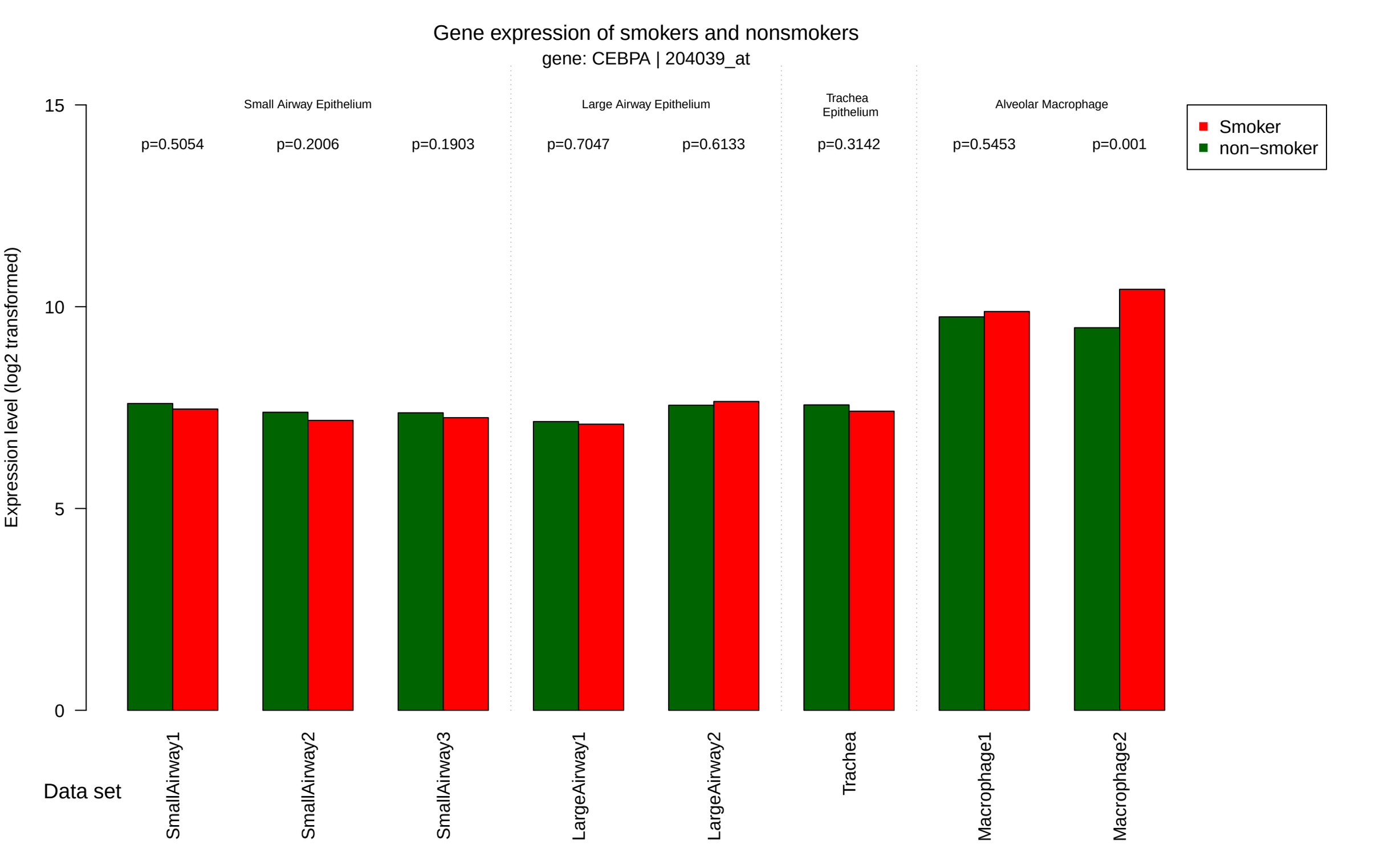
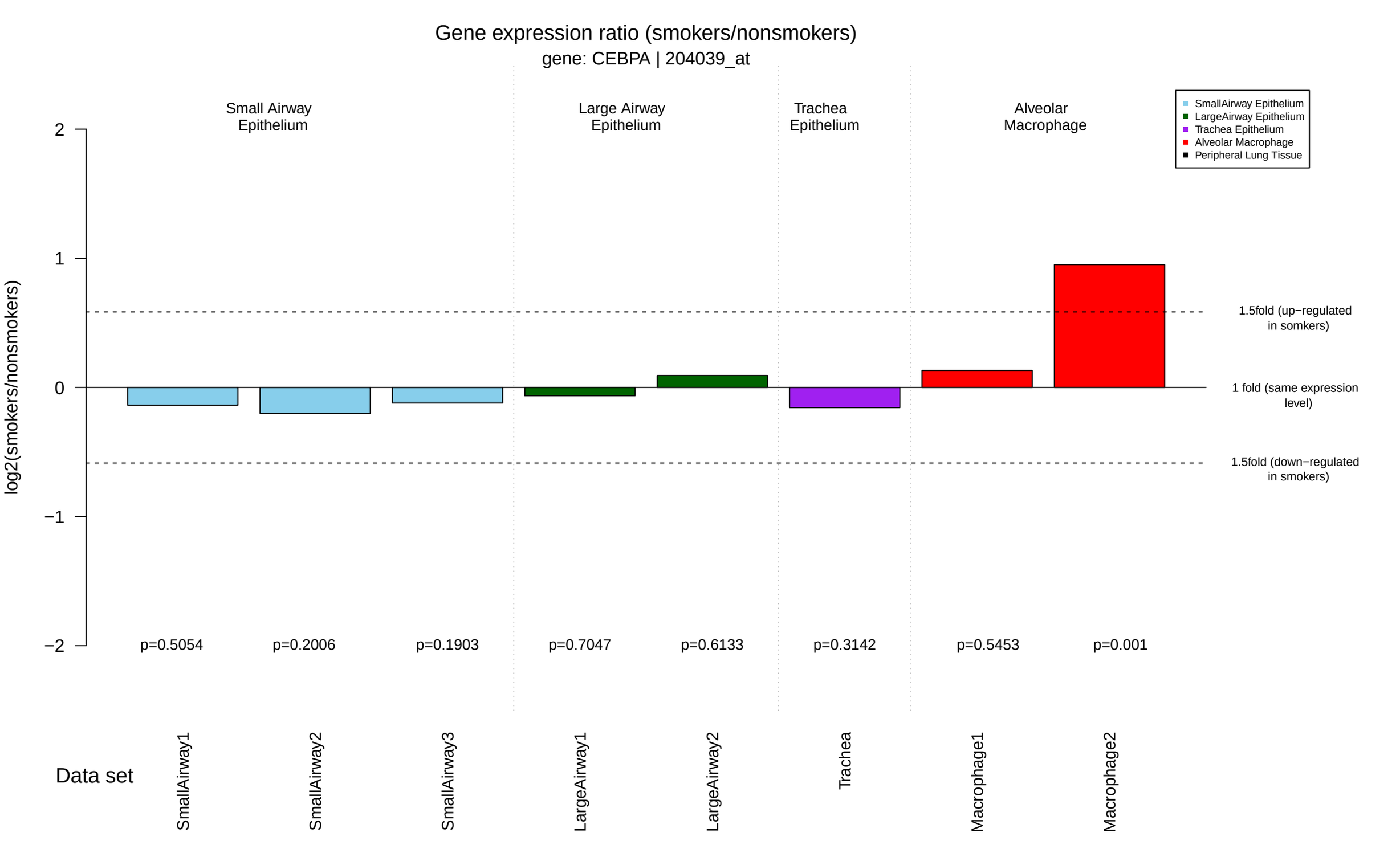
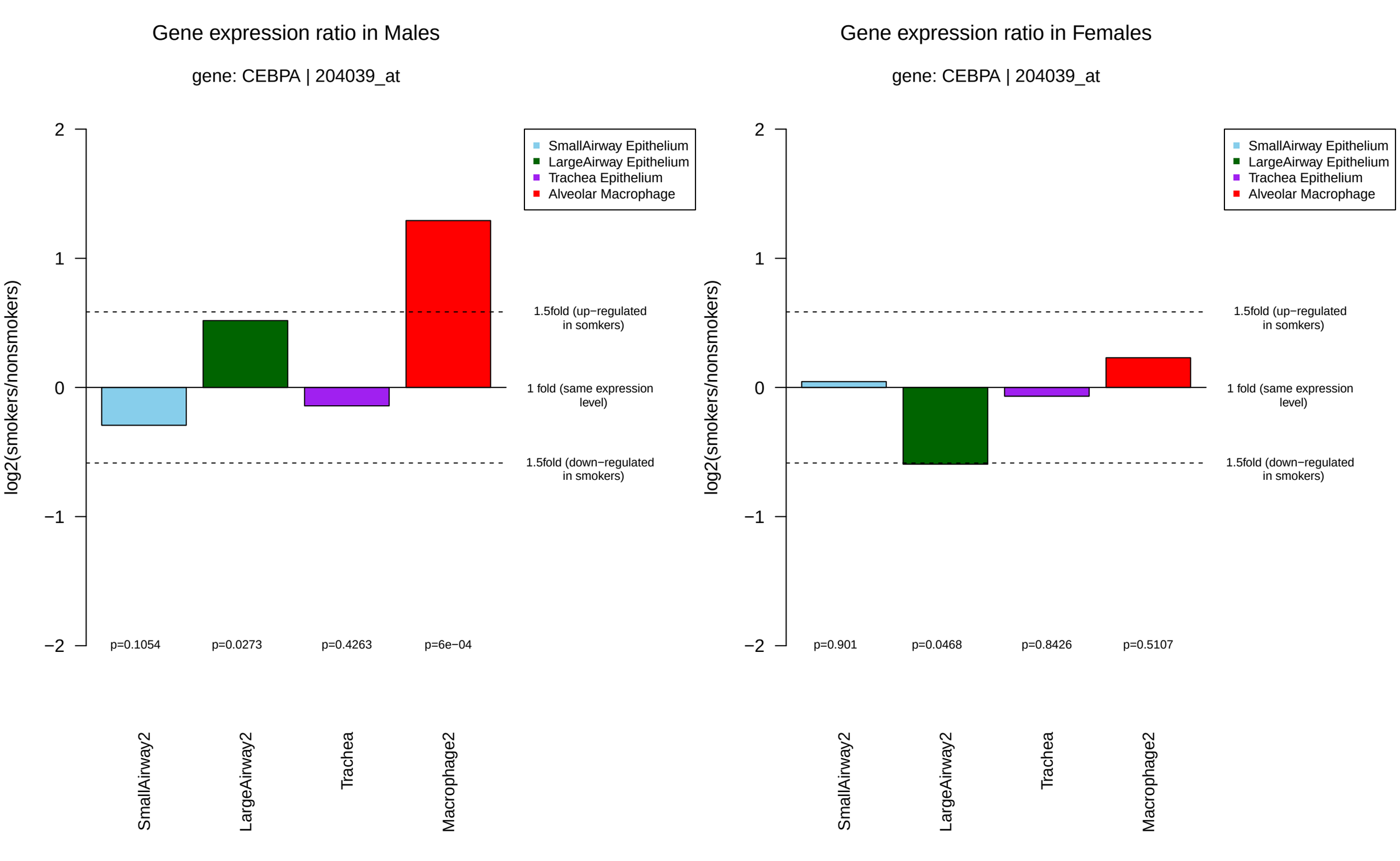
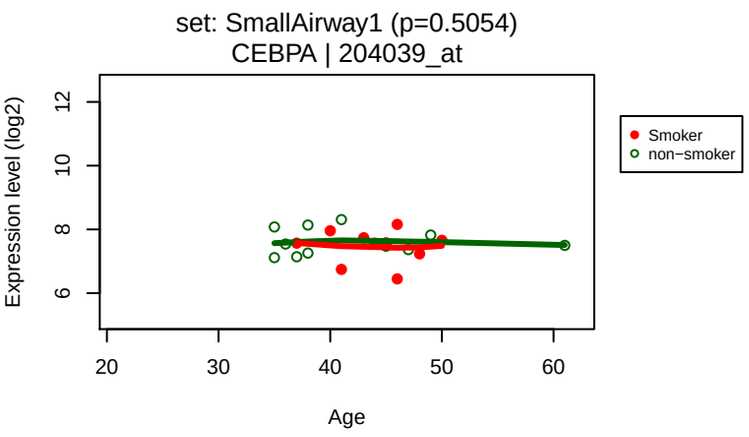
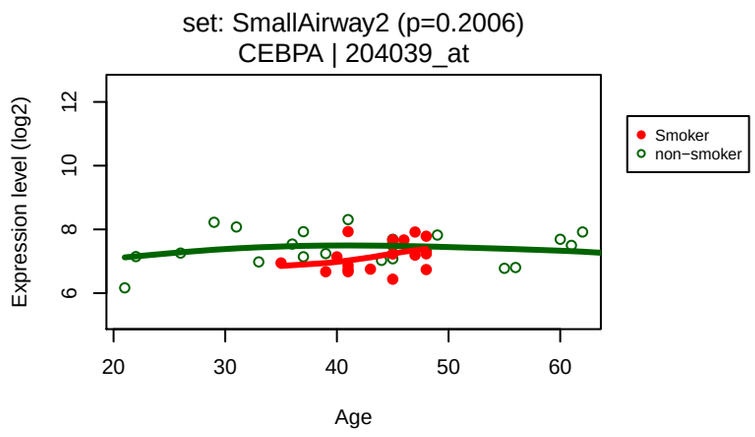
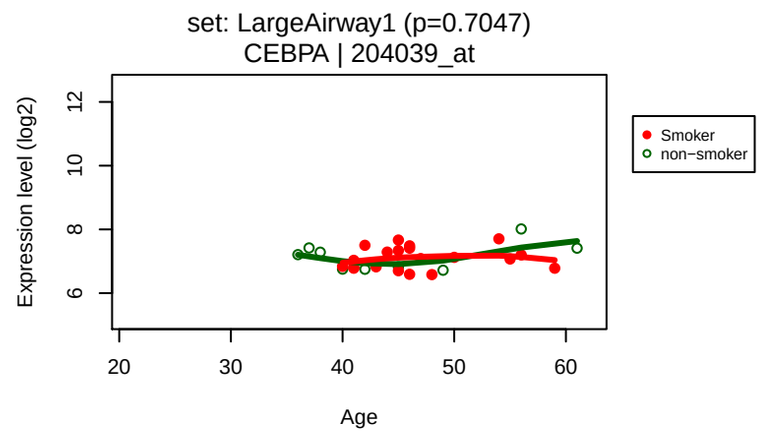

Gene expression of smokers and nonsmokers
gene: CEBPA | 204039_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5054
p=0.2006
p=0.1903
p=0.7047
p=0.6133
p=0.3142
p=0.5453
p=0.001
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

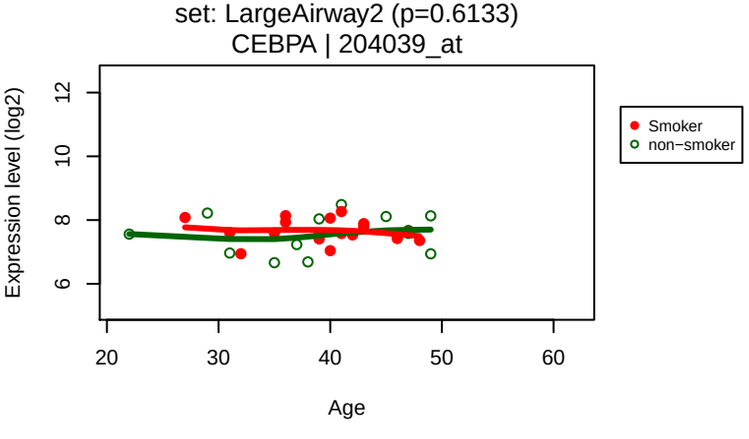

Gene expression ratio (smokers/nonsmokers)
gene: CEBPA | 204039_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5054
p=0.2006
p=0.1903
p=0.7047
p=0.6133
p=0.3142
p=0.5453
p=0.001
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

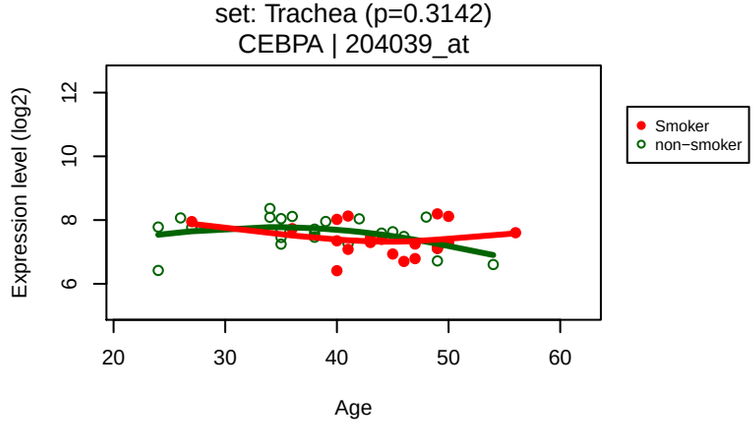

Gene expression ratio in Males
gene: CEBPA | 204039_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1054
p=0.0273
p=0.4263
p=6e−04
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CEBPA | 204039_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.901
p=0.0468
p=0.8426
p=0.5107
SmallAirway2
LargeAirway2
Trachea
Macrophage2

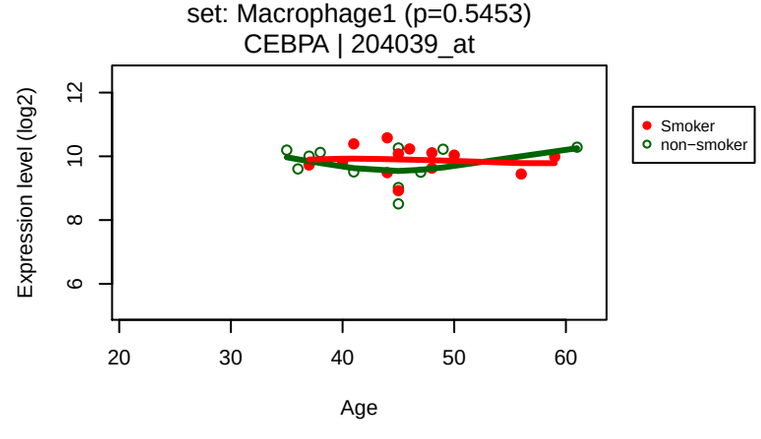

set: SmallAirway1 (p=0.5054)
CEBPA | 204039_at
Expression level (log2)
Age
Smoker
non−smoker

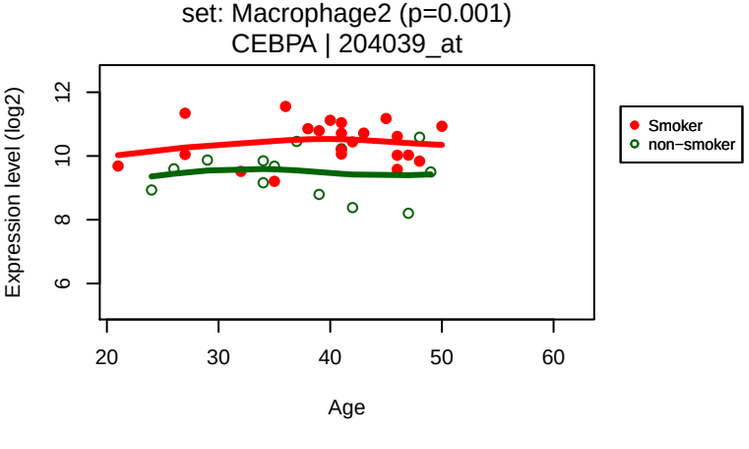

set: SmallAirway2 (p=0.2006)
CEBPA | 204039_at
Expression level (log2)
Age
Smoker
non−smoker

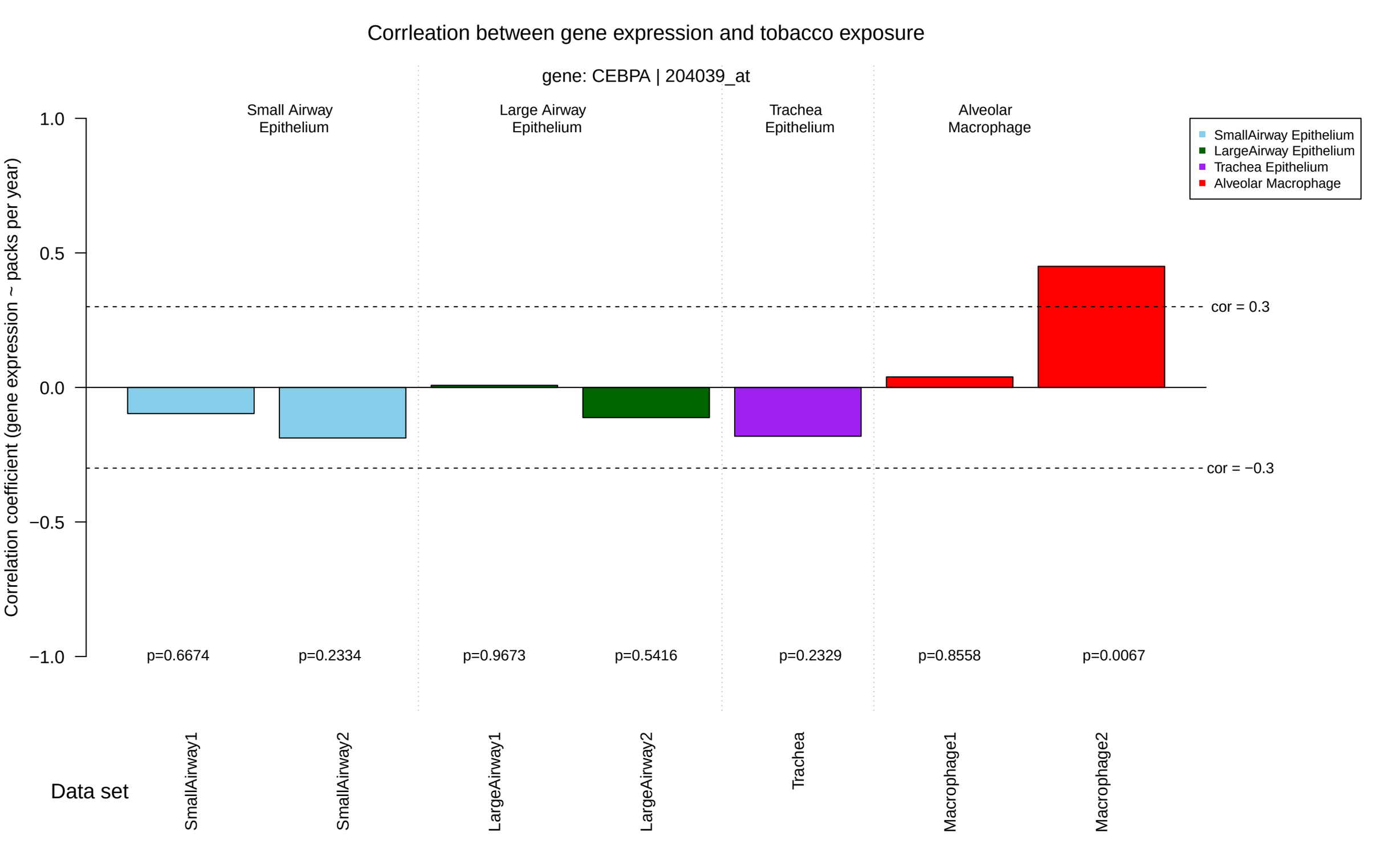

set: LargeAirway1 (p=0.7047)
CEBPA | 204039_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6133)
CEBPA | 204039_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3142)
CEBPA | 204039_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5453)
CEBPA | 204039_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.001)
CEBPA | 204039_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CEBPA | 204039_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.6674
p=0.2334
p=0.9673
p=0.5416
p=0.2329
p=0.8558
p=0.0067
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2